

## Step1: Build and push Docker image to GAR

1. Create a fresh new github repo private  
Add and push Dockerfile, req.txt, iris\_fastapi.py, model.joblib into main
2. Setup Artifact Registry for docker images  
Go to artifact registry  
Create repository  
Name:iris  
Region: us-central1(Iowa)  
Go inside that repository and copy the path. You need to paste it inplace of \*  
Docker build -t \*/iris-api:latest  
Docker push \*/iris-api
3. Create service account in GCP  
Go to IAM and Admin-> service accounts  
Create service account  
Service account id: GitHub-actions done  
  
Copy the service account email and go to artifact registry  
select our repository(iris)  
Show info panel on right top corner  
Add principle  
New principals: paste the copied email from service account  
Role: Artifact registry administrator/writer  
Save  
  
Go back to service account and create a JSON key  
It will be saved in your computer  
Open that file copy the entire text use it in place of #
4. Inside GitHub secrets and variables-> action  
Create new repository secrets  
1.  
name: GOOGLE\_APPLICATION\_CREDENTIALS  
Value: #  
2.  
name: GOOGLE\_PROJECT  
Value: precise-braid-474114-m3/ PROJECT-ID
5. Create docker-image.yaml inside .github/workflows => Build and Push Docker Images

## Step 2: Deploy using Google Kubernetes Engine from GitHub Actions

1. Create cluster in GKE (you can create from command line as well)  
go to Kubernetes engine->cluster->create  
keep everything as it is name **autopilot-cluster-1**

2. Create a deployment.yaml file (not inside workflows)
3. Service account needs to have permission not only for Artifact Registry but also for Kubernetes engine

IAM and admin->IAM->grant access->

New principals: paste the copied email from service account

Role: kubernetes engine developer

Save

4. Create new build-deploy.yaml inside .github/workflows or modify the previous one

## **TROUBLESHOOT**

kubectl apply -f service.yaml => Create or update the Service in the cluster using this YAML

kubectl logs deploy/iris-api => Shows the live logs from all pods in the iris-api deployment.

kubectl get svc iris-api => Shows the current details of the iris-api Service